learningtower

R-CMD-check passing

(https://github.com/kevinwang09/learningtower/actions/workflows/R-CMD-check.yaml)

The goal of `learningtower` is to provide a user-friendly R package to provide easy access to a subset of variables from PISA data collected from the OECD (https://www.oecd.org/pisa/data/). Version 1.1.0 of this package provides the data for the years 2000 - 2022. The survey data is published every three years. This is an excellent real world dataset for data exploring, data visualising and statistical computations.

Links

View on CRAN (https://cloud.r-project.org/package=learningtower)

Browse source code (https://github.com/kevinwang09/learningtower/

Report a bug (https://github.com/kevinwang09/learningtower/

License

Full license (LICENSE.html)

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Citation

Citing learningtower (authors.html#citation)

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What is the PISA dataset?

The Programme for International Student Assessment (PISA) is an international assessment measuring student performance in reading, mathematical and scientific literacy.

PISA assesses the extent to which 15-year-old students have acquired some of the knowledge and skills that are essential for full participation in society, and how well they are prepared for lifelong learning in the areas of reading, mathematical and scientific literacy.

In 2022, PISA involved 79 countries and 600,000+ students worldwide.

Read more about the Programme here (https://www.oecd.org/en/about/programmes/pisa.html).

Installation

You can install the `learningtower` package from CRAN (https://CRAN.R-project.org) with:

```
install.packages("learningtower")
```

To install the development version of `learningtower` from GitHub (https://github.com/) use:

```
devtools::install_github("kevinwang09/learningtower")
```

Data Description

The `learningtower` gives access to a subset of variables from PISA data originally collected and are available from OECD (https://www.oecd.org/pisa/data/), collected on a three year basis.

The `learningtower` package contains mainly three datasets:

- `student`
- `school`
- `countrycode`

This provides us with information about the students scores in mathematics, reading and science, their school details, and which country they are from. The data provided in this package is a cleaned version of the full published PISA organisation, with reproducible code available in this repository (https://github.com/kevinwang09/learningtower_masonry).

The number of entries for the `student` and `school` data are shown below.

Year	Number of Students	Number of Schools
2000	127,236	8,526
2003	276,165	10,274
2006	398,750	14,365
2009	515,958	18,641
2012	480,174	18,139
2015	519,334	17,908
2018	612,004	21,903
2022	613,744	21,629

Student Dataset

The `student` dataset comprises of the scores from the triennial testing of 15-year-olds worldwide. In addition, this dataset contains interesting information on their parents qualifications, family wealth, gender, and possession of computers, internet, cars, books, rooms, desks, and similar other variables.

The full dataset is approximately 50MB in size, which is much larger than the CRAN's allowed package size limit. As the result, the package itself only includes a random 50 rows from the 38 OECD countries (https://en.wikipedia.org/wiki/OECD#Member_countries), for each of the survey years.
i.e. `student_subset_2000` , `student_subset_2003` etc.

The `student` subset dataset can be loaded easily. See `?student` for detailed information on the measured variables.

```
library(learningtower)

data(student_subset_2018)
dim(student_subset_2018)
#> [1] 1900   22
```

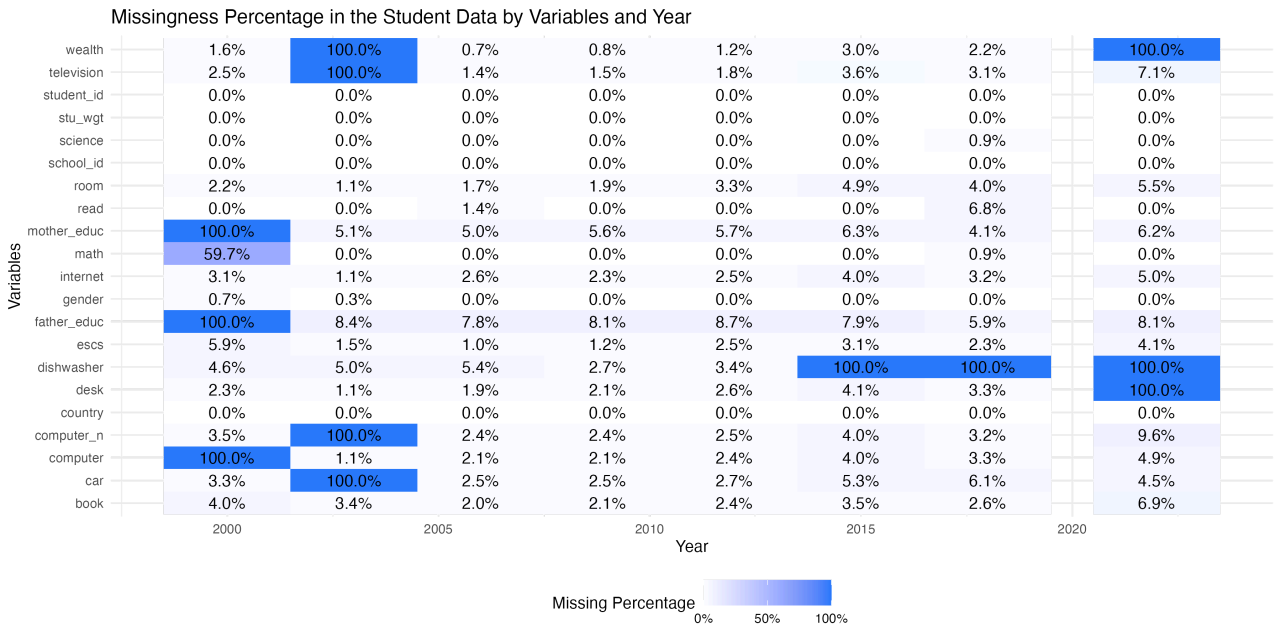
The entire `student` data can be downloaded using the `load_student` function.

```
#load the entire student data for a single year
student_data_2018 <- load_student(2018)

#load the entire student data for two of the years (2012, 2018)
student_data_2012_2018 <- load_student(c(2012, 2018))

#load the entire student data
student_data_all <- load_student("all")
```

Note that because of changing data specification over the survery years, not all variables were measured consistently across the years.



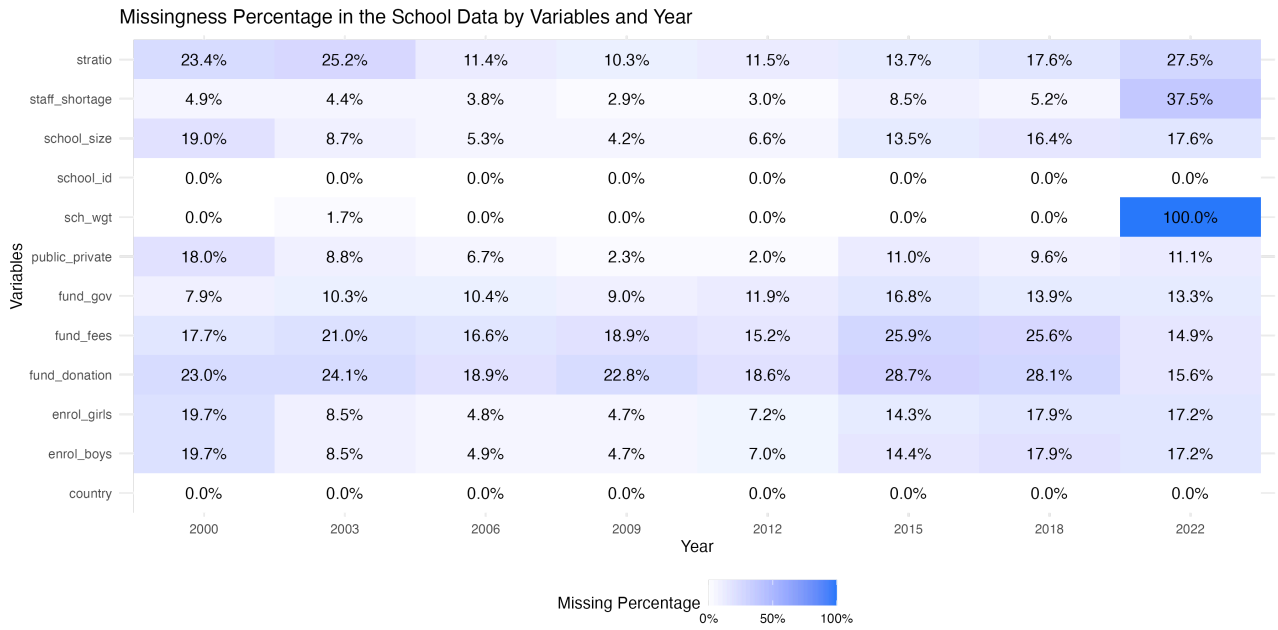
School Dataset

The `school` dataset comprises school weight and other information such as the funding distribution of the schools, whether the school is private or public, the enrollment of boys and girls, the school size, and similar other characteristics of interest of different schools these 15-year-olds attend throughout the world.

- The school subset dataset can be loaded as follows

```
# loading the school data
data(school)
```

See `?school` for more information on the different variables present in the the school dataset.



Countrycode Dataset

The countrycode dataset contains mapping of the country ISO code to the country name (https://www.oecd.org/content/dam/oecd/en/about/programmes/edu/pisa/publications/technical-report/PISA2015_TechRep_Final.pdf). More information on the participating countries can be found here (https://www.oecd.org/en/about/programmes/pisa/pisa-participants.html).

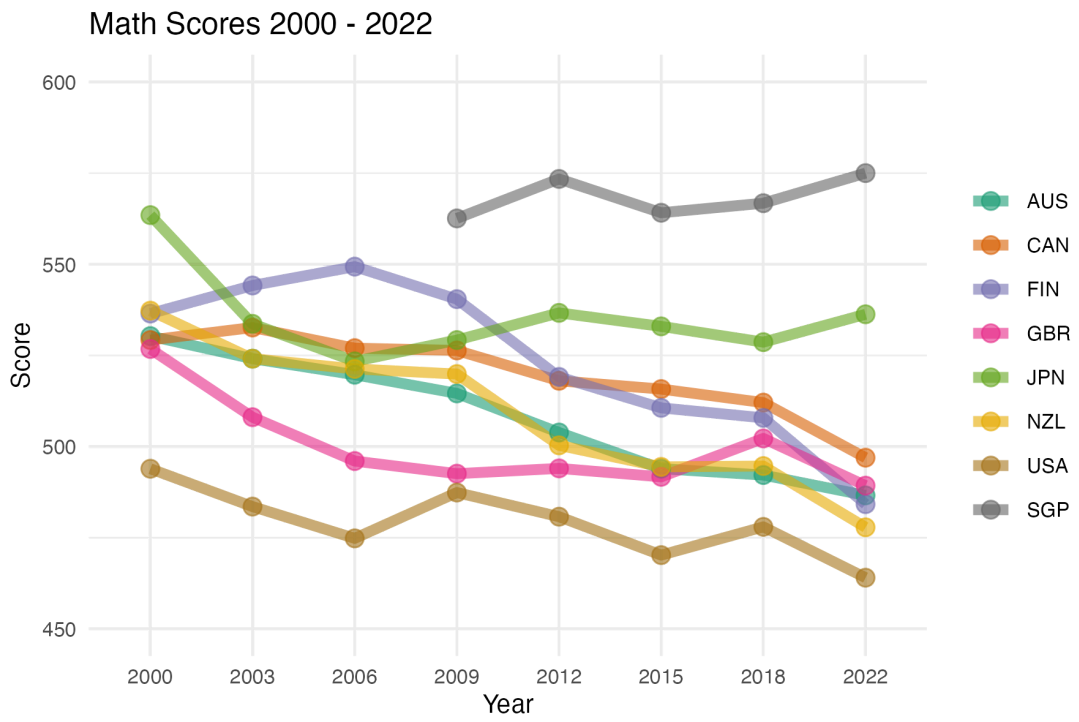
```
# loading the countrycode data
data(countrycode)
head(countrycode)
#> # A tibble: 6 × 2
#>   country country_name
#>   <chr>   <chr>
#> 1 AZE     Azerbaijan
#> 2 ARG     Argentina
#> 3 AUS     Australia
#> 4 AUT     Austria
#> 5 BEL     Belgium
#> 6 BRA     Brazil
```

▶ Notes on countries

See `?countrycode` for more detailed information on the countries that participated in the PISA experiment.

Exploring the data

In the plot shown below, shows the weighted mean of mathematics scores of these 15 year old students for a few selected countries over the available years.



- Similarly, you can find more code examples and data visualizations for exploring `learningtower` through our vignettes and articles
- Further data exploration can be found in our articles exploring temporal trends here (https://kevinwang09.github.io/learningtower/articles/articles/exploring_time.html).

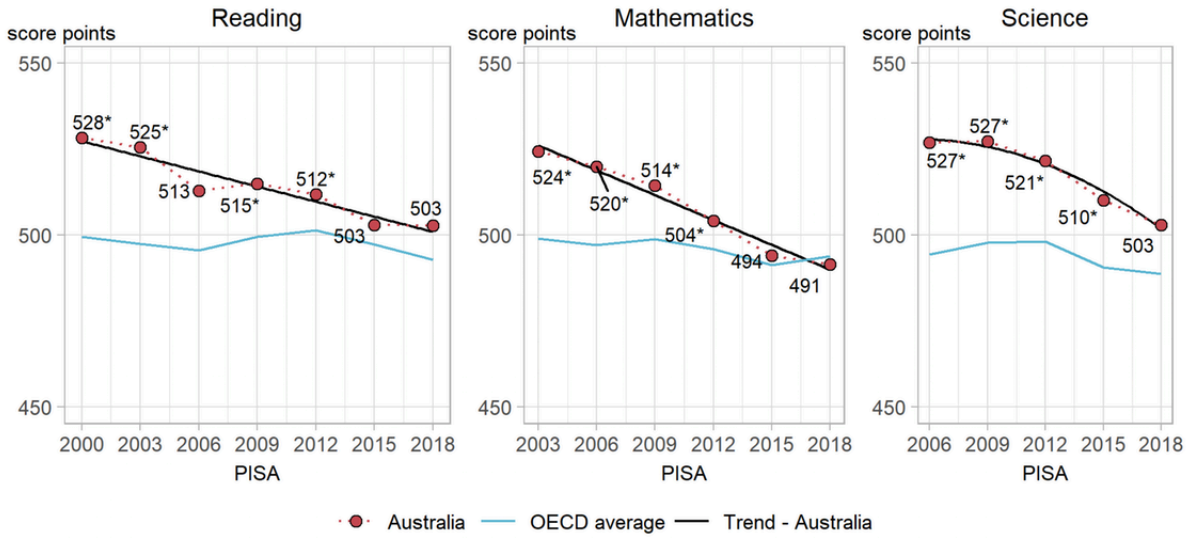
Citation

To cite the `learningtower` package, please use:

```
citation("learningtower")
#> To cite package 'learningtower' in publications use:
#>
#>   Wang K, Yacobellis P, Siregar E, Romanes S, Fitter K, Dalla Riva G,
#>   Cook D, Tierney N, Dingorkar P, Sai Subramanian S, Chen G (2024).
#>   _learningtower: OECD PISA Datasets from 2000-2022 in an Easy-to-Use
#>   Format_. R package version 1.1.0,
#>   https://github.com/kevinwang09/learningtower,
#>   <https://kevinwang09.github.io/learningtower/>.
#>
#> A BibTeX entry for LaTeX users is
#>
#>   @Manual{,
#>     title = {learningtower: OECD PISA Datasets from 2000-2022 in an Easy-to-Use Form
at},
#>     author = {Kevin Wang and Paul Yacobellis and Erika Siregar and Sarah Romanes and
Kim Fitter and Giulio Valentino {Dalla Riva} and Dianne Cook and Nick Tierney and Priya
Dingorkar and Shabarish {Sai Subramanian} and Guan Ru Chen},
#>     note = {R package version 1.1.0, https://github.com/kevinwang09/learningtower},
#>     url = {https://kevinwang09.github.io/learningtower/},
#>     year = {2024},
#>   }
```

Motivation for `learningtower`

- The PISA 2018 results were released on 3 December 2019. This led to wringing of hands in the Australian press, with titles of stories like Vital Signs: Australia's slipping student scores will lead to greater income inequality (https://theconversation.com/vital-signs-australias-slipping-student-scores-will-lead-to-greater-income-inequality-128301) and In China, Nicholas studied maths 20 hours a week. In Australia, it's three (https://www.smh.com.au/education/in-china-nicholas-studied-maths-20-hours-a-week-in-australia-it-s-three-20191203-p53ggv.html).



- Australia's neighbours, New Zealand and Indonesia, are also worrying: New Zealand top-end in OECD's latest PISA report but drop in achievements 'worrying' (https://www.stuff.co.nz/national/education/117890945/new-zealand-topend-in-oecds-latest-pisa-report-but-drop-in-achievements-worrying), Not even mediocre? Indonesian students score low in math, reading, science: PISA report (https://www.thejakartapost.com/news/2019/12/04/not-even-mediocre-indonesian-students-score-low-in-math-reading-science-pisa-report.html).
- The data from this survey and all of the surveys conducted since the first collection in 2000, is publicly available. We decided to have made a more convenient subset of the data available in a new R package, called `learningtower`

Acknowledgement

The work to make the data available is the effort of several researchers from Australia, New Zealand and Indonesia, conducted as part of the ROpenSci OzUnconf (https://ozunconf19.ropensci.org) held in Sydney, Dec 11-13, 2019.